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2025

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section–A

Objective type Questions

Compulsory

1. Choose the correct answer of the following multiple choice questions : 2×10=20

(i) Feedback from users is mainly obtaned :

A. At the end of the project

B. During every phase

C. Before requirement analysis

D. During design phase only

(ii) Which of the following is a major drawback of the Waterfall model ?

A. Easy to manage

B. Clear documentation

C. Difficult to accommodate changes

D. Well-defined stages

(iii) Which of the following is a major advantage of the incremental model ?

A. Simple documentation

B. Easy error detection in early stages

C. No need of planning

D. No maintenance required

(iv) In the incremental model, the first increment usually delivers :

A. Complete system

B. Core/basic functionality

C. Documentation only

D. Final product

(v) Each increment in the Incremental Model adds:

A. New hardware

B. New requirements only

C. Additional functionality

D. New users

(vi) A prototype is:

A. Final software product

B. Complete system design

C. Working model of the system

D. Testing document

(vii) In black box testing, the tester focuses on:

A. Internal code structure

B. Program logic

C. Input and output behavior

D. Flow of control

(viii) In white box testing, the tester requires knowledge of:

A. User requirements

B. Internal code structure

C. User interface

D. System documentation only

(ix) Which of the following is the first level of testing?

A. Integration testing

B. System testing

C. Unit testing

D. Acceptance testing

(x) CMMI stands for :

A. Capability Management Model Integration

B. Capability Maturity Model Integration

C. Computer Maturity Model Integration

D. Common Maturity Model Interface

Section–B

Short Answer type Questions

Answer any **four** questions of the following : 5×4=20

2. Write the steps of prototyping model.

3. Write disadvantages of Waterfall Model.

4. What are functional and non-functional requirements in software engineering?

5. Why is Software Requirements Specification needed?

6. What is meant by software, and what are its key characteristics?

7. What are the objectives of CMMI.

8. What is software engineering, and why is it important in the software development process?

Section–C

Long Answer type Questions

Answer any **two** questions of the following : 15×2=30

9. Explain the incremental Model in software engineering and describe its phases.

10. Explain the characteristics of SRS.

11. What is software metrics? Explain the types of software metrics.

12. Explain the maturity levels of CMMI.

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